

Yijun Liu

yijunliu523.github.io · yijun6@illinois.edu · Urbana, IL, USA

Research Interest

I design and study AI writing- and feedback-support systems grounded in human expertise, combining interactive system-building with human-AI alignment evaluation. My work spans human-computer interaction, human-AI collaboration, and human-centered NLP.

Education

- 2024 - Present **University of Illinois Urbana-Champaign**, Urbana, IL
Ph.D. Candidate in Computer Science
Advisor: Tal August
- 2020 - 2024 **Emory University**, Atlanta, GA
B.S. in Computer Science (honors); B.S. in Quantitative Sciences in Sociology
Honor Thesis Advisor: Chinmay Kulkarni

Peer-Reviewed Publication

- [C4] From Crafting Text to Crafting Thought: Grounding Intelligent Writing Support to Writing Center Pedagogy
Yijun Liu, John Gallagher, Sarah Serman, and Tal August
CHI 2026
*Preliminary Version: In2Writing Workshop at NAACL 2025 *Best Paper Award**
- [C3] Needling Through the Threads: A Visualization Tool for Navigating Threaded Online Discussions
Yijun Liu, Frederick Choi, and Eshwar Chandrasekharan
CHI 2026
- [C2] Unmasking Dunning-Kruger Effect in Visual Reasoning and Visual Data Analysis
Mengyu Chen, Yijun Liu, and Emily Wall
2024 IEEE Visualization Conference (VIS'24)
- [J1] Predicting Atrial Fibrillation Ablation Outcomes: A Machine Learning Approach Leveraging a Large Administrative Claims Database
Yijun Liu, Mustapha Oloko-Oba, Kathryn Wood, Michael S. Lloyd, Joyce C. Ho, Vicki Stover Hertzberg
JMIR Cardio 2025
- [C1] Redesigning Mobile Crowd Sensing's Selection Logic to Ensure Participant's Diversity
Jiahui Jin, Yijun Liu, Ting Li
IPCCC 2023

Preprints and Workshop Papers

- [P4] FOXGLOVE: Understanding Goal-Oriented and Anchored Writing Feedback from Experts and LLMs on Argumentative Essays
Yijun Liu, Yifan Song, John Gallagher, Sarah Serman, Tal August
2026
- [P3] Creative Reading: Scaffolding Reading for Transformation
Sophia Liu, Sarah Abowitz, Yijun Liu, Sarah Serman, Shm Garanganao Almeda, Max Kreminski
2026

- [P2] Breaking the Curse of Knowledge: Designing Personalized Jargon Support for Real-Time Online Meetings
Yifan Song, [Yijun Liu](#), Wing Yee Au, Hon Yung Wong, Brian P Bailey, Tal August 2026
- [W1] StarTales: Imagining Interactive Storyreading with Large Language Models
[Yijun Liu](#)
STAR Workshop @ CHI 2026
- [P1] Machine Learning-Based Identification of High-Risk Patterns in Atrial Fibrillation Ablation Outcomes
Mustapha Oloko-Oba, [Yijun Liu](#), Kathryn Wood, Michael S. Lloyd, Joyce C. Ho, Vicki Stover Hertzberg 2024

Award & Recognition

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| 2025 | In2Writing Best Paper Award |
| 2024 | Emory Outstanding CS Student of the Year
Phi Beta Kappa |
| 2023 | Google CS Research Mentorship Recipient |
| 2022 | SIGKDD Student Travel Award (\$800)
CRA-WP Student Travel Award (\$2,000)
Emory Quantitative Science Distinction Award
ASA DataFest @ Emory Best Insight Award |
| 2021 | ASA Fall Data Challenge Best Undergraduate Overall Award |

Research Experience

Evaluating AI and Expert Writing Feedback [P4]

Advisor: Tal August

- Built FOXGLOVE, a benchmark dataset of 696 sentence-anchored, goal-labeled feedback annotations across 69 argumentative essays for evaluating AI writing-support systems.
- Engineered a multi-model evaluation pipeline benchmarking four frontier LLMs (GPT-5.2, Claude Sonnet 4.5, Llama 3.3, and Qwen3-Next) against human annotators on goal alignment and positional accuracy.
- Designed a 7-dimensional expert rating framework and software, as well as implemented alignment metrics to quantify human-LLM divergence.

Designing Non-Directive AI Writing Feedback [C4]

Advisor: Tal August

- Designed and built *Writor*, an AI writing-feedback tool informed by interviews with 10 writing tutors.
- Conducted a mixed-methods expert evaluation with 30 educators (survey + interview) assessing pedagogical alignment and feedback quality; AI-skeptical educators showed a strong preference for non-directive feedback.

Visualizing Threaded Activity in Online Discussion for Moderators [C3]

Advisor: Eshwar Chandrasekharan

- Led design and development of *Needle*, an interactive visualization system using Next.js/D3.js to summarize Reddit threads by activity, toxicity, and engagement.
- Conducted qualitative user study with 10 Reddit moderators to assess usability and impact of *Needle* on moderation workflow; found that *Needle* provides a practical solution to maintain contextual understanding when navigating threaded conversations.

Enhancing Collaborative Web App Inclusivity via LLMs [Honor Thesis]

Advisor: Chinmay Kulkarni

- Investigated whether LLMs can lower the barrier to accessible web development by automatically retrofitting inclusivity features into existing front-end code.
- Built an iterative generate-validate-merge pipeline that produced functioning, inclusivity-enhanced React pages from existing code.

Predicting Atrial Fibrillation Ablation Success with ML [J1]

Advisors: Vicki Hertzberg & Joyce Ho

- Generated refined datasets from 100GB+ healthcare data in SAS, applying time-series transformation and normalization to gather patients' medical histories.
- Led development of ML models, including logistic regression, SVMs, decision trees, random forests, GBDTs, and neural networks; employed k-fold, Monte Carlo cross-validation, grid search, and various feature selection techniques (filter, wrapper & embedded) alongside with PCA.
- Advanced performance by developing specialized subgroup models surpassing existing benchmarks.

Teaching Experience

2026 SP	CS 498: Human-LLM Interaction , Teaching Assistant, UIUC
2025 FA	CS 465: User Interface Design , Teaching Assistant, UIUC
2022 FA	CS 253: Data Structure and Algorithms , Teaching Assistant, Emory
2021 FA & SP	Emory Oxford Writing Center , Lead Writing Consultant

Academic Service

Web Developer

2024	<u>AAAI Workshop: LLMs4Bio</u>
2022	<u>SIGKDD</u>

Student Volunteer

2022	<u>SIGKDD</u>
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Reviewer

2025	CHI 2026 (Special Recognition for outstanding reviews)
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Skill

Technical Skills	Machine Learning & NLP (scikit-learn, PyTorch, TensorFlow, Keras, LLM APIs, prompt engineering, RAG, LLM evaluation), Front-End (HTML, CSS/SASS, TypeScript, JavaScript, React.js, Next.js), Back-End (Flask, Node.js, REST APIs), Database (MySQL, PostgreSQL, MongoDB, Firebase), Cloud Services (AWS, GCP, Heroku), Statistical Analysis (Python, NumPy, Pandas, SciPy, R, SAS), Visualization (Matplotlib, D3.js)
Design	Prototyping, Figma, Adobe Photoshop/InDesign/Illustrator
Research Methods	Interviews, Survey Design, Usability Testing, Experimental Design, Hypothesis Testing, Qualitative & Thematic Analysis, Mixed-Methods Analysis